

Entropy

May 14, 2026

Exercise 1.

Prove the following (in)equalities:

1. Chain rule: $H(X, Y) = H(X) + H(Y|X)$
2. Subadditivity: $H(X, Y) \leq H(X) + H(Y)$
3. Submodularity: $H(X, Y, Z) + H(X) \leq H(X, Y) + H(X, Z)$

Solution 1.

We start by proving the Chain rule:

$$\begin{aligned} H(X, Y) &= \sum_{x,y} \mathbb{P}(X = x, Y = y) \log_2(\mathbb{P}(X = x, Y = y)) \\ &= \sum_x \underbrace{\left(\sum_y \mathbb{P}(Y = y|X = x) \right)}_{=1} \mathbb{P}(X = x) \log_2 \mathbb{P}(X = x) \\ &\quad + \sum_x \mathbb{P}(X = x) \left(\sum_y \mathbb{P}(Y = y|X = x) \log_2 \mathbb{P}(Y = y|X = x) \right) \\ &= H(X) + 1 \cdot H(Y|X) \end{aligned}$$

For the Subadditivity, consider the convex function $f(t) = \log_2(1/t)$. We have

$$\begin{aligned} H(X) + H(Y) - H(X, Y) &= \sum_{x,y} \mathbb{P}(X = x, Y = y) (-\log_2(\mathbb{P}(X = x)) - \log_2(\mathbb{P}(Y = y)) + \log_2(\mathbb{P}(X = x, Y = y))) \\ &= \sum_{x,y} \mathbb{P}(X = x, Y = y) \log_2 \left(\frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)\mathbb{P}(Y = y)} \right) \\ &= \sum_{x,y} \mathbb{P}(X = x, Y = y) f \left(\frac{\mathbb{P}(X = x)\mathbb{P}(Y = y)}{\mathbb{P}(X = x, Y = y)} \right) \\ &\geq f \left(\sum_{x,y} \mathbb{P}(X = x, Y = y) \frac{\mathbb{P}(X = x)\mathbb{P}(Y = y)}{\mathbb{P}(X = x, Y = y)} \right) \\ &= f(1) = 0 \end{aligned}$$

$$H(Y|X) = \sum_x \mathbb{P}(X = x) \sum_y \mathbb{P}(Y = y|X = x) \log_2(\mathbb{P}(Y = y|X = x))$$

Finally, to prove submodularity, we do not need to unpack the definition of entropy and can simply use the various bounds we have seen so far:

$$\begin{aligned} H(X, Y, Z) + H(X) &= H(X, Y) + H(Z|X, Y) + H(X) && \text{by Chain rule} \\ &\leq H(X, Y) + H(Z|X) + H(X) && \text{by dropping the conditional} \\ &\leq H(X, Y) + H(X, Z) && \text{by Chain rule} \end{aligned}$$

Exercise 2.

Let $m \in \mathbb{N}$ and $S_m = \sum_{k=1}^{10} k^{-m}$. Let X_m be a random variable such that for every $k \in [10]$, $\mathbb{P}(X_m = k) = \frac{1}{S_m k^m}$. Without any calculations, what can you say about the sequence $(H(X_m))_{m \in \mathbb{N}}$?

Solution 2.

For $m = 0$, X_0 is a uniform random variable so $H(X_0) = \log_2(10)$. As m grows, $\mathbb{P}(X_m = 1)$ increases and tends to 1, X_m becomes more and more predictable, so $H(X_m)$ should decrease and tend to 0.

Exercise 3.

Let $p \in [0, 1]$ and $n \in \mathbb{N}^+$ such that pn is integral. Prove that $\frac{2^{nH(p)}}{8\sqrt{p(1-p)}} \leq \binom{n}{pn}$ using Stirling's formula:

$$\sqrt{2\pi n} \left(\frac{n}{e}\right)^n < n! < \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$

Solution 3.

Denote $k = pn$. As we have seen in the lecture, $2^{nH(p)} = p^{-k}(1-p)^{n-k}$. We have

$$\begin{aligned} \binom{n}{k} &= \frac{n!}{k!(n-k)!} \\ &\geq \frac{\sqrt{n}}{\sqrt{2\pi k(n-k)} e^{\frac{1}{12k} + \frac{1}{12(n-k)}}} \left(\frac{n}{k}\right)^k \left(\frac{n}{n-k}\right)^{n-k} \\ &\geq \frac{1}{8\sqrt{p(1-p)}} 2^{nH(p)} \end{aligned}$$

1 Counting Latin squares

A Latin square of size n is an $n \times n$ array $(s_{ij})_{i,j \in [n]}$ such that each line and each column contains every number from $[n]$.

Exercise 4. Counting Latin squares using Brégman's result on the permanent

Brégman proved that for every matrix $A \in \{0, 1\}^{n \times n}$, the permanent of A is at most $\prod_i (d_i!)^{1/d_i}$, where d_i is the number of ones in the i -th line. Using this result, prove that the number of Latin squares of size n is at most $n! \prod_{k=1}^{n-1} (k!)^{n/k}$.

Hint : Assuming that you already filled the k first lines of the Latin square, count the possible extensions

Solution 4.

An $k \times n$ Latin rectangle is a $k \times n$ array such that each line contains all colours from $[n]$ and each column contains k distinct colours of $[n]$. For every k , for every $l \times n$ Latin rectangle L , we bound the number of $(k+1) \times n$ Latin rectangles that extend L . Each cell in the $(k+1)$ -th line can contain any colour that did not appear in the same column in one of the previous rows. Let A be the matrix of $\{0, 1\}^{n \times n}$ whose entry a_{ij} is 1 if colour j does not appear in the i -th column of L . The number of $(k+1) \times n$ Latin rectangles that extend L is the permanent of A . As A has exactly $(n-k)$ ones per line, its permanent is at most $((n-k)!)^{n/(n-k)}$. As a result, the number of $n \times n$ Latin squares is at most $n! \prod_{k=1}^{n-1} (k!)^{n/k}$.

★ Exercise 5. Counting Latin squares using entropy

The goal of this exercise is to prove that the number of Latin squares of size n is at most $\left(\frac{n^2}{e^{2+o(1)}}\right)^{n^2}$

To do so, consider a random uniform Latin square and reveal its entries in a random order, like we did for matchings. We will use (and admit) that $\int_0^1 \log_2(1 + (n-1)z^2) dz = \log_2(n) - 2 + o(1)$.

Hint : To sample the random order, pick uniform independent reals in $[0, 1]$

★ **Solution 5.**

The method is a classic entropy method to upper bound the number of solutions (here Latin squares):

1. Express the entropy of a random uniform Latin square X . By uniformity, the number of Latin squares is $2^{H(X)}$.
2. Reveal the entries of X in a random order. By permuting the expectations over the order and the entries, we first compute the expectation over the order with a fixed Latin square (which is often easier), before averaging on the Latin square.
3. To compute this first expectation, we often use convexity arguments.

Let X be a random uniform Latin square of size n . Since X is uniform, the number of Latin squares of size n will be exactly counted by $2^{H(X)}$. Let $X_{i,j}$ be the random variable corresponding to the entry in the intersection of the i -th line and the j -th column.

Given any fixed ordering τ of the cells, the Chain rule gives

$$H(X) = \sum_{(i,j) \in [n] \times [n]} H(X_{i,j} | X_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j)) \quad (1)$$

Let τ be a random ordering of $[n] \times [n]$, i.e. a random ordering τ of the cells of the Latin square. More precisely, it will be convenient to sample τ using for an array Z of uniform independent random reals $Z = (Z_{i,j}) \in [0,1]^{n \times n}$, and order the cells by increasing values of $Z_{i,j}$. By summing (??) over all possible values of z and normalizing,

$$\begin{aligned} H(X) &= \sum_z \mathbb{P}(Z = z) \left[\sum_{(i,j) \in [n] \times [n]} H(X_{i,j} | X_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j)) \right] \\ &= \mathbb{E}_z \left[\sum_{(i,j) \in [n] \times [n]} H(X_{i,j} | X_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j)) \right] \\ &= \sum_{(i,j) \in [n] \times [n]} \mathbb{E}_z [\mathbb{E}_x [H(X_{i,j} | X_{i',j'} = x_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j))]] \\ &= \sum_{(i,j) \in [n] \times [n]} \mathbb{E}_x [\mathbb{E}_z [H(X_{i,j} | X_{i',j'} = x_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j))]] \quad \text{because } X \text{ is discrete} \end{aligned}$$

The entry $X_{i,j}$ can take all possible values that are not already in a cell $(i',j') <_{\tau(Z)} (i,j)$ lying in the same line or column as (i,j) . We say that a colour $s \in [n]$ is *available* for (i,j) if $s \notin \{X_{i',j'} : (i',j') \text{ and } (i,j) \text{ are in the same row or column, and } (i',j') <_{\tau(Z)} (i,j)\}$. The number $N_{X,\tau(Z)}(i,j)$ of available colours for (i,j) varies between 1 (because X is a Latin square) and n . For a fixed Latin square x and a random ordering $\tau(Z)$, as the conditional entropy of $X_{i,j}$ is at most the logarithm of its support, we have

$$\begin{aligned} \mathbb{E}_z [H(X_{i,j} | X_{i',j'} = x_{i',j'} \text{ for all cells } (i',j') \text{ before } (i,j))] &\leq \mathbb{E}_z [\log_2(N_{X,\tau(Z)})] \\ &\leq \mathbb{E}_{z_{i,j}} [\log_2(\mathbb{E}_{\{z_{i',j'} : (i',j') \neq (i,j)\}} [N_{X,\tau(Z)}])] \end{aligned}$$

because logarithm is a convex function. Now for any fixed x and $z_{i,j}$, let us compute the expected number of available colours for (i,j) . The colour $s = x_{i,j}$ does not appear in the other cells of the i -th row or the j -th column, because x is a Latin square, so $x_{i,j}$ is always available. For any other colour s , there exists two cells (i',j) and (i,j') such that $x_{i',j} = x_{i,j'} = s$. If any of them appears before (i,j) , then s is not available at (i,j) . So

$$\begin{aligned} \mathbb{P}(s \text{ is available for } (i,j)) &= \mathbb{P}(Z_{i',j} > z_{i,j} \text{ and } Z_{i,j'} > z_{i,j}) \\ &= \mathbb{P}(Z_{i',j} > z_{i,j}) \mathbb{P}(Z_{i,j'} > z_{i,j}) \quad \text{by independence of } Z_{i',j} \text{ and } Z_{i,j'} \\ &= (1 - z_{i,j})^2 \end{aligned}$$

Hence, $\mathbb{E}_{z_{i,j}} [\log_2(\mathbb{E}_{\{z_{i',j'} : (i',j') \neq (i,j)\}} [N_{X,\tau(Z)}])] = \int_0^1 \log_2(1 + (n-1)(1-z)^2) dz \leq \log_2 n - 2 + o(1)$. Thus, $H(X) \leq n^2(\log_2 n - 2 + o(1))$ and the number of Latin squares is at most $2^{H(X)} \leq ((1 + o(1)) \frac{n}{e^2})^{n^2}$